

Thermodynamic Constraints and Energy Budget Framework for Nonequilibrium Electromagnetic Systems: Implications for Emergent Vacuum Response Theory (EVRT)

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Abstract

This work develops a thermodynamically consistent framework for analyzing driven, nonequilibrium electromagnetic systems of the type explored in Emergent Vacuum Response Theory (EVRT). Rather than proposing violations of energy conservation or free-energy generation, the study establishes a device-level energy-budget formalism grounded in classical electrodynamics, equilibrium thermodynamic potentials, and nonequilibrium loss accounting. The framework integrates Helmholtz free energy for confined electromagnetic structures and Gibbs free energy considerations for environmental interactions and open-system artifact channels. A generalized power balance equation is derived for high-voltage asymmetric capacitor-coil systems, including mechanical output, dielectric polarization, radiative losses, thermal pathways, and parasitic effects. The formulation defines a residual power term as an uncertainty-bounded diagnostic quantity rather than evidence of new physics. An illustrative numerical estimate is included to show how stored-field energy, torsion-balance output, and measurement uncertainty can be compared on a common scale. This approach provides a falsifiable, constraint-driven pathway for investigating whether structured electromagnetic systems can redistribute supplied energy into measurable mechanical effects within an uncertainty-bounded experimental framework beyond conventional expectations, without invoking violations of thermodynamic laws.

Keywords: nonequilibrium thermodynamics; Helmholtz free energy; Gibbs free energy; electromagnetic stress; energy budget; torsion balance; EVRT; residual force; artifact discrimination

1 Introduction

Recent investigations into structured electromagnetic systems have explored whether strongly asymmetric, driven field configurations may produce measurable force or stress-energy effects under nonequilibrium conditions. Such systems may include high-voltage capacitor geometries, resonant coil structures, dielectric boundaries, and precision force measurement apparatuses. Because these configurations involve strong fields, environmental coupling, and multiple loss pathways, they are susceptible to misinterpretation unless analyzed through a rigorous energy-accounting framework.

Emergent Vacuum Response Theory (EVRT) is proposed as a constrained phenomenological program for investigating whether structured nonequilibrium electromagnetic systems can produce measurable stress redistribution or mechanical response beyond conventional expectations. The present work does not assert free-energy generation, reactionless propulsion, or violations of conservation laws. Instead, it develops a thermodynamic consistency framework intended to protect

the interpretation of EVRT-class experiments from overclaiming and to define what would count as a meaningful residual signal.

The central objective is to formulate a complete power balance for an EVRT tabletop device, identify the roles of Helmholtz and Gibbs free energy, and establish consistency conditions that any proposed anomalous effect must satisfy. In this view, a genuine experimental anomaly would not be interpreted as energy creation, but as a statistically significant and reproducible redistribution of supplied electromagnetic free energy into a measurable mechanical or stress-energy channel not captured by a specified conventional model. The framework therefore treats any residual signal as a metrological problem first and a physical interpretation problem second.

2 Thermodynamic Foundations

2.1 Helmholtz Free Energy for Confined Electromagnetic Systems

For a system maintained at approximately constant temperature and volume, the appropriate thermodynamic potential is the Helmholtz free energy,

$$A = U - TS, \quad (1)$$

where A is the Helmholtz free energy, U is internal energy, T is absolute temperature, and S is entropy. For a closed or approximately confined electromagnetic device, this form is especially relevant because the device volume is fixed and the dominant energy reservoirs are electromagnetic fields, dielectric polarization, inductive storage, and thermal dissipation.

The maximum extractable useful work under constant-temperature and constant-volume conditions is bounded by

$$W_{\max} \leq -\Delta A. \quad (2)$$

In rate form, a conservative bound on mechanical output is

$$P_{\text{mech}} \leq -\frac{dA}{dt}, \quad (3)$$

provided the system boundary and stored-energy terms are properly defined. This bound applies under quasi-static conditions and well-defined system boundaries. Strongly driven nonequilibrium regimes may require generalized formulations that explicitly account for time-dependent field structure and entropy production.

For an EVRT-class capacitor-coil device, the stored internal energy may be approximated by

$$U_{\text{stored}} = \frac{1}{2}C(t)V(t)^2 + \frac{1}{2}L(t)I(t)^2 + U_{\text{pol}}(t), \quad (4)$$

where $C(t)$ is the effective capacitance, $V(t)$ is the applied voltage, $L(t)$ is the effective inductance, $I(t)$ is current, and $U_{\text{pol}}(t)$ represents dielectric polarization energy.

2.2 Gibbs Free Energy for Environmental Coupling

For open-system interactions at approximately constant temperature and pressure, the Gibbs free energy is

$$G = H - TS, \quad (5)$$

where H is enthalpy. The change in Gibbs free energy is often expressed as

$$\Delta G = \Delta H - T\Delta S. \quad (6)$$

A process with $\Delta G < 0$ is thermodynamically favorable under the stated conditions, while $\Delta G = 0$ corresponds to equilibrium and $\Delta G > 0$ requires external input.

In EVRT experiments, Gibbs free energy is most relevant to environmental and artifact channels, including corona discharge, ion wind, electrohydrodynamic flow, thermal convection, humidity-dependent leakage, and gas-phase ionization. These effects are not secondary details; they can dominate apparent force measurements if not isolated. Therefore, the device core is most naturally described through Helmholtz free energy, while the surrounding environment and open-system artifacts are constrained by Gibbs free energy.

3 Representative EVRT Device

The representative EVRT tabletop device considered here consists of a driven asymmetric electromagnetic structure mounted on a sensitive torsion balance or equivalent force sensor. The device may include:

- high-voltage asymmetric capacitor plates or electrodes;
- a lightweight dielectric support structure;
- magnet wire or coil elements for driven magnetic-field structure;
- optional resonant or pulsed operation;
- electrical isolation and shielding;
- angular displacement measurement through optical, capacitive, or interferometric readout.

System Boundary Definition.

For the purposes of energy accounting, the system boundary must be explicitly defined for each experiment. At minimum, this boundary includes the electromagnetic device, its immediate electrical connections, and any structures within significant electromagnetic coupling range. Failure to properly define the system boundary can result in misattribution of stored energy, loss pathways, or mechanical coupling, and therefore directly affects the interpretation of the residual term.

The central measured quantity is mechanical displacement, torque, or force. The critical scientific question is not whether the device can produce motion under power, but whether any observed mechanical response remains after subtracting all known stored-energy changes, environmental effects, and parasitic loss channels.

4 Full Energy Budget Equation

A complete power balance for an EVRT-class device can be written as

$$P_{\text{in}} = \frac{dU_{\text{stored}}}{dt} + P_{\text{mech}} + P_{\text{loss}} + P_{\text{res}}, \quad (7)$$

where P_{in} is the total input power, dU_{stored}/dt is the rate of change of stored electromagnetic and polarization energy, P_{mech} is mechanical output power, P_{loss} is the sum of identified loss channels, and P_{res} is a residual term.

For a device with both high-voltage and coil-drive contributions,

$$P_{\text{in}} = P_{\text{HV}} + P_{\text{drive}}. \quad (8)$$

The stored-energy term is

$$U_{\text{stored}} = \frac{1}{2}C(t)V(t)^2 + \frac{1}{2}L(t)I(t)^2 + U_{\text{pol}}(t). \quad (9)$$

Thus, the device-level balance becomes

$$\begin{aligned} P_{\text{HV}} + P_{\text{drive}} = & \frac{d}{dt} \left[\frac{1}{2}C(t)V(t)^2 + \frac{1}{2}L(t)I(t)^2 + U_{\text{pol}}(t) \right] + P_{\text{mech}} \\ & + P_{\text{Joule}} + P_{\text{dielectric}} + P_{\text{corona}} + P_{\text{RF}} + P_{\text{vib}} + P_{\text{thermal}} + P_{\text{leak}} + P_{\text{res}}. \end{aligned} \quad (10)$$

The known-loss aggregate is therefore

$$P_{\text{loss}} = P_{\text{Joule}} + P_{\text{dielectric}} + P_{\text{corona}} + P_{\text{RF}} + P_{\text{vib}} + P_{\text{thermal}} + P_{\text{leak}}. \quad (11)$$

5 Mechanical Output on a Torsion Balance

For a torsion-balance measurement, the mechanical output can be written as

$$P_{\text{mech}} = \tau\omega, \quad (12)$$

where τ is torque and ω is angular velocity. If the torsion constant is κ and angular deflection is θ , then

$$\tau = \kappa\theta. \quad (13)$$

The mechanical power becomes

$$P_{\text{mech}} = \kappa\theta\dot{\theta}. \quad (14)$$

For static or quasi-static deflection, the stored torsional energy is

$$E_{\text{mech}} = \frac{1}{2}\kappa\theta^2. \quad (15)$$

If the arm length from pivot to device center of mass is r , an equivalent force estimate is

$$F = \frac{\tau}{r} = \frac{\kappa\theta}{r}. \quad (16)$$

These relations allow a measured angular response to be converted into a mechanical-energy term that can be compared against input power, stored-field variation, and loss estimates.

6 Illustrative Numerical Energy-Budget Estimate

The following estimate is not presented as an experimental result. It is included only to demonstrate how an EVRT-class measurement can be placed on a common energy scale before any anomalous interpretation is considered. Consider a representative high-voltage asymmetric capacitor–coil device with an effective capacitance

$$C = 10 \text{ pF} = 1.0 \times 10^{-11} \text{ F} \quad (17)$$

and applied voltage

$$V = 10 \text{ kV} = 1.0 \times 10^4 \text{ V}. \quad (18)$$

The capacitive stored energy is then

$$U_C = \frac{1}{2}CV^2 = \frac{1}{2} (1.0 \times 10^{-11}) (1.0 \times 10^4)^2 = 5.0 \times 10^{-4} \text{ J}. \quad (19)$$

For a torsion balance with torsion constant

$$\kappa = 1.0 \times 10^{-7} \text{ N m rad}^{-1} \quad (20)$$

and quasi-static angular displacement

$$\theta = 1.0 \times 10^{-3} \text{ rad}, \quad (21)$$

the stored torsional energy is

$$E_{\text{mech}} = \frac{1}{2}\kappa\theta^2 = \frac{1}{2} (1.0 \times 10^{-7}) (1.0 \times 10^{-3})^2 = 5.0 \times 10^{-14} \text{ J}. \quad (22)$$

For an arm length

$$r = 0.10 \text{ m}, \quad (23)$$

the corresponding equivalent force is

$$F = \frac{\kappa\theta}{r} = \frac{(1.0 \times 10^{-7}) (1.0 \times 10^{-3})}{0.10} = 1.0 \times 10^{-9} \text{ N}. \quad (24)$$

This nanonewton-scale force estimate is many orders of magnitude smaller than the stored capacitive energy scale. Therefore, a measured deflection of this magnitude cannot be evaluated by displacement alone; it must be compared against the complete time-dependent budget of input power, stored-field variation, mechanical damping, thermal drift, electrostatic coupling, and environmental loss channels.

For example, if the electrical input power during a measurement interval were

$$P_{\text{in}} = 1.0 \text{ W} \quad (25)$$

and a residual after accounting were reported as

$$P_{\text{res}} = 2.0 \times 10^{-6} \text{ W}, \quad (26)$$

then the residual would represent only

$$\frac{P_{\text{res}}}{P_{\text{in}}} = 2.0 \times 10^{-6} \quad (27)$$

of the applied input power. Such a small fraction would be scientifically interesting only if its combined uncertainty were substantially smaller than the residual and if it survived null tests, polarity reversal, dummy-load substitution, thermal controls, vacuum operation, and shielding.

This disparity underscores that any claimed anomalous force must be interpreted as a small perturbation superimposed on a substantially larger energy flow. Consequently, the detection of a meaningful residual requires that systematic uncertainties, environmental couplings, and unmodeled loss channels be constrained to levels well below the inferred signal magnitude.

7 Residual Power and Interpretation

Time Averaging and Measurement Interval.

All power and residual terms are implicitly defined over a finite measurement interval. In practice, P_{in} , dU_{stored}/dt , P_{mech} , and P_{loss} should be evaluated as time-averaged quantities over a window chosen to suppress transient charging effects, dielectric relaxation, switching artifacts, and instrumentation noise. The choice of averaging interval must be reported alongside any residual measurement.

The residual term is defined by rearranging Eq. 7:

$$P_{\text{res}} = P_{\text{in}} - \left(\frac{dU_{\text{stored}}}{dt} + P_{\text{mech}} + P_{\text{loss}} \right). \quad (28)$$

A result satisfying

$$P_{\text{res}} = 0 \pm \sigma_{P_{\text{res}}} \quad (29)$$

within measurement uncertainty indicates that the device is fully explained by known energy storage, mechanical output, and identified losses.

Here $\sigma_{P_{\text{res}}}$ denotes the combined standard uncertainty associated with the residual estimate.

A persistent nonzero residual should be evaluated through the significance ratio

$$Z_{\text{res}} = \frac{|P_{\text{res}}|}{\sigma_{P_{\text{res}}}}. \quad (30)$$

A residual should not be interpreted as physically meaningful unless it is reproducible, sign-stable under appropriate reversals, and large compared with its combined uncertainty.

A statistically significant residual would not by itself demonstrate new physics. It would instead indicate a hierarchy of possibilities:

1. calibration, instrumentation, synchronization, or data-reduction error;
2. an incorrectly defined system boundary or omitted stored-energy term;
3. an unmeasured conventional loss or coupling pathway;
4. a reproducible candidate structured anomaly requiring further controlled investigation.

Only the fourth interpretation becomes plausible after repeated exclusion of conventional artifacts, including ion wind, thermal drift, electrostatic coupling, RF leakage, Lorentz-force coupling to external conductors, vibration, acoustic forces, and charging of nearby surfaces.

8 Thermodynamic Consistency Conditions

Any EVRT-class interpretation must satisfy the following conditions.

Null Configuration Requirement.

A valid experimental protocol must include null configurations in which the expected residual is identically zero by construction. Examples include symmetric electrode configurations, disconnected or dummy electrical loads, and polarity-reversal conditions. Agreement with null expectations provides a necessary baseline for interpreting any nonzero residual.

8.1 Energy Conservation

The complete system must obey

$$P_{\text{in}} = \frac{dU_{\text{stored}}}{dt} + P_{\text{mech}} + P_{\text{loss}} + P_{\text{res}}, \quad (31)$$

with P_{res} treated as an experimental diagnostic, not as evidence of energy creation.

8.2 Entropy Production

The total entropy production of the device plus environment must satisfy

$$\frac{dS_{\text{total}}}{dt} \geq 0. \quad (32)$$

Loss terms such as Joule heating, dielectric relaxation, RF emission, corona, and mechanical damping contribute to positive entropy production.

8.3 Work Bound

Mechanical output must remain bounded by available free energy:

$$P_{\text{mech}} \leq -\frac{dA}{dt} \quad (33)$$

for the confined device core, or by the relevant open-system free-energy balance when environmental coupling is significant.

8.4 No Free-Energy Claim

A thermodynamically consistent EVRT interpretation must not require sustained mechanical output with no input energy, no decrease in stored energy, and no compensating entropy production. Such a result would conflict with established thermodynamics unless independently reproduced under extraordinary experimental scrutiny.

8.5 Uncertainty Propagation and Residual Criterion

The residual power should be reported with a combined uncertainty rather than as a single deterministic value. If the input, stored-energy, mechanical, and loss terms are treated as independent first-order contributions, then

$$\sigma_{P_{\text{res}}}^2 \approx \sigma_{P_{\text{in}}}^2 + \sigma_U^2 + \sigma_{P_{\text{mech}}}^2 + \sigma_{P_{\text{loss}}}^2, \quad (34)$$

where $\sigma_{\dot{U}}$ is the uncertainty in dU_{stored}/dt . If correlations exist among measured quantities, the covariance terms must be retained:

$$\sigma_{P_{\text{res}}}^2 = \sum_i \left(\frac{\partial P_{\text{res}}}{\partial x_i} \right)^2 \sigma_{x_i}^2 + 2 \sum_{i < j} \left(\frac{\partial P_{\text{res}}}{\partial x_i} \right) \left(\frac{\partial P_{\text{res}}}{\partial x_j} \right) \text{Cov}(x_i, x_j). \quad (35)$$

A conservative reporting criterion is therefore

$$P_{\text{res}} = 0 \pm \sigma_{P_{\text{res}}} \quad (36)$$

for null-consistent operation, and

$$|P_{\text{res}}| > n\sigma_{P_{\text{res}}} \quad (37)$$

for a provisional residual, where n should be chosen before data collection. In exploratory work, a threshold such as $n = 3$ may be used as a screening criterion, while stronger claims require independent replication, blind controls, and systematic error studies.

9 Artifact Channels and Control Strategy

These requirements are consistent with established practices in precision force metrology, including torsion-balance and microforce measurement systems, where rigorous control of systematic error and uncertainty propagation is essential for interpreting small signals.

Table 1 summarizes major artifact channels and corresponding controls.

Table 1: Representative artifact channels in EVRT-class experiments.

Artifact Channel	Mechanism	Control Strategy
Ion wind / EHD flow	Force from ionized air and charge transport	Vacuum operation, pressure sweep, polarity reversal
Corona discharge	Gas ionization and asymmetric plasma forces	Dark testing, acoustic monitoring, current spikes, reduced pressure study
Thermal drift	Expansion, convection, radiometric forces	Thermal imaging, dummy heaters, symmetric thermal controls
Electrostatic coupling	Attraction to nearby charged or grounded objects	Shielding, guard electrodes, distance scaling
RF leakage	Momentum transfer or induced currents in surroundings	Faraday shielding, spectrum monitoring, load substitution
Lorentz-force coupling	Interaction with external wires or ambient magnetic fields	Twisted-pair routing, battery isolation, field mapping
Vibration / acoustic coupling	Mechanical excitation from power supplies or discharge	Isolation stage, accelerometers, off-device dummy load
Dielectric relaxation	Slow charge migration and delayed mechanical response	Material comparison, discharge protocol, humidity control

10 Energy and Entropy Flow Diagrams

Figure 1 summarizes the device-level energy budget for a driven EVRT-class experiment. The residual term is shown as a diagnostic quantity rather than as evidence of energy creation.

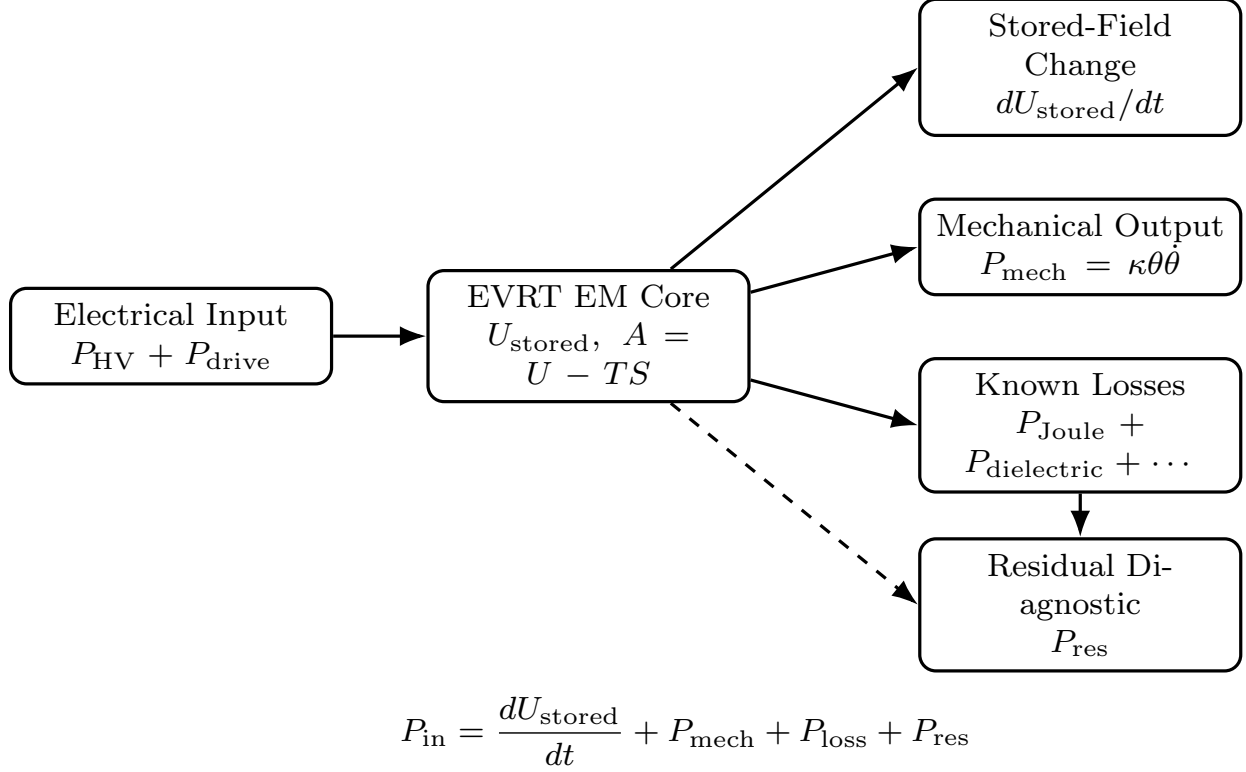
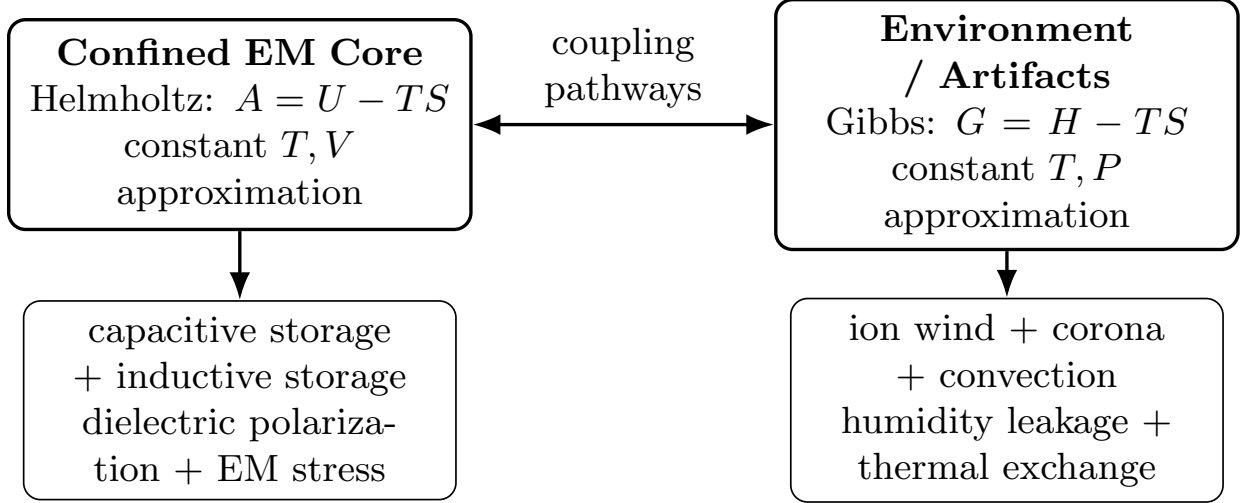


Figure 1: Energy-budget flow for a driven EVRT-class electromagnetic device. The diagram expresses the experimental accounting structure used to compare input power, stored-field variation, mechanical output, known loss channels, and the residual diagnostic term.

Figure 2 separates the two thermodynamic domains used in this framework. The confined electromagnetic core is treated primarily through Helmholtz free energy, while open environmental channels and artifacts are constrained through Gibbs free energy and entropy production.



Global consistency: $dS_{\text{total}}/dt \geq 0$ and $P_{\text{mech}} \leq P_{\text{in}}$

Figure 2: Thermodynamic domain separation for EVRT-class experiments. The device core is modeled through a Helmholtz free-energy perspective, while environmental interaction channels such as corona, ion wind, convection, and leakage are governed by Gibbs-type open-system constraints.

11 Implications for the EVRT Program

This framework reframes EVRT as a thermodynamically constrained investigation of structured electromagnetic energy redistribution. Under this interpretation, EVRT does not require energy creation or violation of conservation laws. Instead, it asks whether nonequilibrium electromagnetic geometry, dielectric structure, and resonant drive conditions can redirect supplied energy into measurable mechanical or stress-energy channels in a way not captured by simplified conventional models.

The residual term P_{res} provides a disciplined target for future experiments, but only when reported together with $\sigma_{P_{\text{res}}}$ and the pre-defined significance criterion used to evaluate it. This requirement forces all claims to pass through energy accounting, loss characterization, entropy production, uncertainty propagation, and environmental control. It is particularly important for high-voltage experiments, where conventional artifacts can easily mimic small forces.

A rigorous EVRT research program should therefore proceed by:

1. defining the system boundary before each experiment;
2. measuring total electrical input power rather than relying only on voltage;
3. quantifying stored capacitive, inductive, and dielectric energy;
4. measuring or bounding all known loss channels;
5. converting mechanical deflection into torque, force, and energy;
6. reporting P_{res} with uncertainty bounds and treating any residual as provisional until reproduced under control conditions.

12 Conclusion

A thermodynamic consistency framework has been developed for EVRT-class nonequilibrium electromagnetic systems. The confined electromagnetic core is most naturally described using Helmholtz free energy, while environmental interactions and artifact channels are constrained by Gibbs free energy and entropy production. A full energy budget equation was derived for a representative high-voltage asymmetric capacitor–coil device mounted on a torsion balance, and an illustrative numerical estimate was provided to place stored-field energy, nanonewton-scale mechanical response, and residual power on a common scale. The resulting uncertainty-bounded residual term provides a falsifiable experimental target while preserving strict adherence to conservation laws.

The proposed framework does not support claims of free-energy generation or thermodynamic violation. Instead, it enables a disciplined investigation of whether structured electromagnetic systems can redistribute supplied energy into measurable mechanical effects beyond conventional expectations. In this form, EVRT becomes a constraint-driven research program grounded in energy accounting, entropy tracking, and precision artifact discrimination.

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Data and Code Availability

No experimental data are reported in this manuscript. The equations and framework are theoretical and intended to guide future experimental design and interpretation.

Conflict of Interest Statement

The author declares no competing financial interests.

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